

JENNY ZHANG

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Computer Science & AI Specialization

September 2021 - May 2026

- **Languages:** Python, TypeScript, JavaScript, Java, C++, C, C#, Bash, SQL, PHP, HTML
- **Tools / Frameworks:** Git, Docker, AWS, React, Vue, Node, Flask, Laravel, MySQL, MongoDB
- **Relevant courses:** Artificial Intelligence, Neural Networks, Machine Learning, Data Structures & Algorithms, Computer Networks, Object-Oriented Programming, Operating Systems, Compilers, Statistics, Linear Algebra

EXPERIENCE

Rippling | Software Engineer Intern, San Francisco

May 2025 – August 2025

- Incoming Software Engineer Intern on the HRIS Product team

ArenaX Labs | Software Developer Intern, Toronto

September 2024 – December 2024

- Engineered the state-action environment for ARC's **reinforcement learning algorithm**, an AI-powered gaming SDK, by **reducing feature cardinality by 15%** in the state space using Principal Component Analysis
- Developed attack-combo mechanics for AI-driven characters using **finite state machines**; implemented action sequences and leveraged **class inheritance** to customize transition functions across states
- Refactored game UI in React and TypeScript, leveraging Jotai for state management to achieve **30% faster data persistence** across component hierarchies, reducing render time and improving gameplay smoothness

Knak | Fullstack Developer Intern, Ottawa

September 2023 – August 2024

- **Streamlined marketing processes by 95%** for enterprise companies through developing the editor platform using Vue, PHP, AWS Lambda, and Docker; allowed clients to create email and landing pages within minutes
- Optimized data retrieval from the MySQL database to dynamically display translation workflow options on the frontend based on customer permissions, enhancing user experience in Knak's AI-driven text translation tool
- Engineered optimized MySQL queries in PHP, **boosting query performance by 30%** and enabling seamless client data migration within seconds during feature deployments

ZeMind Game Studios | Game Developer Intern, Toronto

May 2023 – August 2023

- Built a virtual reality (VR) application in Unity/C# for Durham Region Victim Services (DRVS), providing three immersive experiences to help youth navigate exploitative situations; nominated for the Webby awards 2024
- Orchestrated the engineering of a dialogue and scene management system using an **observer design pattern**

Royal Bank of Canada | Software Developer Intern, Toronto

May 2022 – August 2022

- Designed and developed a Flask application **to test 60% of RBC's site actions** with Python, Pytest and Selenium; adopted by team of 10 developers, **reducing CI pipeline test workflow duration by 80%**

PROJECTS

🔗 Neural Network for Image Classification | Python, Scikit-learn, Numpy, Matplotlib

- Implemented backpropagation from scratch with various optimization algorithms including SGD and AdaGrad

🔗 Landing Page Generator | JavaScript, React, Node, MongoDB, Cohere LLM API, OpenAI API

- Link platform that provides a codeless editor to create personalized domains for users in under 2 minutes

🔗 Emulator for Game Boy | C, CMake, SDL

- Game Boy CPU core emulator, effectively translating C code into the SM83 processor's instruction set

🔗 Multiplayer Tetris | C++, XMinig Server

- Tetris game using OOP principles to implement the MVC architecture through graphical/text display observers